

# Homework 2~6

## Homework 2

### Problem 2.1

(1)证明:

$$\binom{r}{r} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r} = \binom{n+1}{r+1}$$

(2)证明:

$$\sum_{k=0}^n \binom{m+k-1}{k} = \binom{n+m}{n}$$

**Answer:**

(1)

$$\binom{r}{r} = \binom{r+1}{r+1} = 1$$

根据

$$\binom{k+1}{l+1} = \binom{k}{l} + \binom{k}{l+1}$$

有

$$\begin{aligned} & \binom{r}{r} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r} \\ &= \binom{r+1}{r+1} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r} \\ &= \binom{r+2}{r+2} + \binom{r+2}{r} + \cdots + \binom{n}{r} \\ &= \cdots \\ &= \binom{n+1}{r+1} \end{aligned}$$

(2)

$$\begin{aligned}
\sum_{k=0}^n \binom{m+k-1}{k} &= \binom{m-1}{0} + \binom{m}{1} + \binom{m+1}{2} + \cdots + \binom{m+n-1}{n} \\
&= \binom{m}{0} + \binom{m}{1} + \binom{m+1}{2} + \cdots + \binom{m+n-1}{n} \\
&= \binom{m+1}{1} + \binom{m+1}{2} + \cdots + \binom{m+n-1}{n} \\
&= \cdots \\
&= \binom{n+m}{m}
\end{aligned}$$

## Problem 2.2

(1)求:  $\sum_{i=1}^n i(i-1)$ ,  $\sum_{i=1}^n i^2$

Hint: Assume Problem 1  $r=2$

(2)求:  $\sum_{i=1}^n i^3$

**Answer:**

(1)

$$\begin{aligned}
\binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \cdots + \binom{n}{2} &= \binom{n+1}{3} = \frac{(n-1)n(n+1)}{6} \\
\binom{1}{1} + \binom{2}{1} + \binom{3}{1} + \cdots + \binom{n}{1} &= \binom{n+1}{2} = \frac{n(n+1)}{2} \\
\therefore \sum_{i=1}^n i(i-1) &= 2 \left( \binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \cdots + \binom{n}{2} \right) = \frac{(n-1)n(n+1)}{3} \\
\therefore \sum_{i=1}^n i^2 &= \sum_{i=1}^n i(i-1) + \sum_{i=1}^n i = \frac{(n-1)n(n+1)}{3} + \frac{n(n+1)}{2} = \frac{n(n+1)(2n+1)}{6}
\end{aligned}$$

(2)

$$\begin{aligned}
\binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \cdots + \binom{n}{3} &= \binom{n+1}{4} = \frac{(n-2)(n-1)n(n+1)}{24} \\
\therefore \sum_{i=3}^n i(i-1)(i-2) &= 6 \left( \binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \cdots + \binom{n}{3} \right) = \frac{(n-2)(n-1)n(n+1)}{4} \\
\sum_{i=3}^n i(i-1)(i-2) &= \sum_{i=1}^n i(i-1)(i-2) = \sum_{i=1}^n i^3 + \sum_{i=1}^n (-3i^2) + \sum_{i=1}^n 2i \\
\therefore \sum_{i=1}^n i^3 &= \frac{(n-2)(n-1)n(n+1)}{4} + 3 \frac{n(n+1)(2n+1)}{6} - 2 \frac{n(n+1)}{2} = \frac{n^2(n+1)^2}{4}
\end{aligned}$$

## Problem 2.3

计算以下表达式，要求结果不能包含求和符号

$$(1) \sum_{k=1}^n \frac{1}{k} \binom{k}{m}$$

$$(2) \sum_{k=0}^n k \binom{k}{m}$$

**Answer:**

(1)

$$\begin{aligned} \frac{1}{k} \binom{k}{m} &= \frac{1}{k} \frac{k!}{(k-m)!m!} = \frac{(k-1)!}{(k-m)!m!} = \frac{(k-1)!}{m(k-m)!(m-1)!} = \frac{1}{m} \binom{k-1}{m-1} \\ \sum_{k=1}^n \frac{1}{k} \binom{k}{m} &= \frac{1}{m} \sum_{k=1}^n \binom{k-1}{m-1} \\ &= \frac{1}{m} \left( \binom{0}{m-1} + \binom{1}{m-1} + \binom{2}{m-1} + \cdots + \binom{n-1}{m-1} \right) \\ &= \frac{1}{m} \left( \binom{m-1}{m-1} + \binom{m}{m-1} + \binom{m+1}{m-1} + \cdots + \binom{n-1}{m-1} \right) \\ &= \frac{1}{m} \binom{n}{m} \end{aligned}$$

(2)

$$\begin{aligned} k \binom{k}{m} &= (k+1) \binom{k}{m} - \binom{k}{m} \\ (k+1) \binom{k}{m} &= \frac{(k+1)k!}{(k-m)!m!} = \frac{(k+1)!}{(k-m)!m!} = \frac{(m+1)(k+1)!}{(k-m)!(m+1)!} = (m+1) \binom{k+1}{m+1} \\ \sum_{k=0}^n k \binom{k}{m} &= (m+1) \sum_{k=0}^n \binom{k+1}{m+1} - \sum_{k=0}^n \binom{k}{m} \\ &= (m+1) \left( \binom{1}{m+1} + \binom{2}{m+1} + \binom{3}{m+1} + \cdots + \binom{n+1}{m+1} \right) \\ &\quad - \left( \binom{0}{m} + \binom{1}{m} + \binom{2}{m} + \cdots + \binom{n}{m} \right) \\ &= (m+1) \left( \binom{m+1}{m+1} + \binom{m+2}{m+1} + \binom{m+3}{m+1} + \cdots + \binom{n+1}{m+1} \right) \\ &\quad - \left( \binom{m}{m} + \binom{m+1}{m} + \binom{m+2}{m} + \cdots + \binom{n}{m} \right) \\ &= (m+1) \binom{n+2}{m+2} - \binom{n+1}{m+1} \end{aligned}$$

## Problem 2.4

求  $f: \{1, 2, 3, \dots, n\} \rightarrow \{1, 2, 3, \dots, n\}$  中非严格增函数的个数

**Answer:**

所求函数的个数与  $g: \{0, 1, 2, \dots, n+1\} \rightarrow \{1, 2, 3, \dots, n\}$ ,  $g(0) = 1, g(n+1) = n$  且  $g$  为非严格增函数的个数相同

令  $k_i = g(i+1) - g(i)$ , 则有  $\sum_{i=0}^n k_i = n - 1$

所以有  $g$  的个数为  $k_0 + k_1 + k_2 + \dots + k_n$  的非负整数解个数

用隔板法易知  $g$  的个数为  $\binom{2n-1}{n}$

## Homework 3

### Problem 3.1

集合  $X_1, X_2, X_3, \dots, X_k$  分别含有  $r_1, r_2, r_3, \dots, r_k$  个元素, 且这  $r_1 + r_2 + r_3 + \dots + r_k$  个元素互不相同. 每个  $X_i$  都有一个全序关系  $\preceq_i$  约束, 且来自不同  $X_i$  的元素之间不可比. 现将  $X$  定义为  $X_1, X_2, X_3, \dots, X_k$  的并集, 并为  $X$  添加全序关系  $\preceq$ , 要求  $\forall i, \preceq_i \subseteq \preceq$ , 求  $\preceq$  的种类数

**Answer:**

相当于将  $r_1 + r_2 + r_3 + \dots + r_k$  个空位分为  $k$  组, 每组分别有  $r_1, r_2, r_3, \dots, r_k$  个空位, 将  $X_1, X_2, X_3, \dots, X_k$  中的元素按其上的全序关系决定的序填充空位即可, 所以  $\preceq$  共有  $\binom{r_1+r_2+r_3+\dots+r_k}{r_1, r_2, r_3, \dots, r_k}$  种

### Problem 3.2

$X$  是一个集合且  $|X| = n$ , 元素为集合的有序对  $\langle A, B \rangle$  满足  $A \subseteq B \subseteq X$ , 求  $\langle A, B \rangle$  的种类数

**Answer:**

设  $|B| = b$ , 则  $A$  共有  $2^b$  种

故  $\langle A, B \rangle$  的种类数为  $\sum_{b=0}^n 2^b \binom{n}{b}$

恰好是  $(1+2)^b$  的二项展开式, 所以共有  $3^b$  种

### Problem 3.3

定义函数的不动点为: 若函数  $\pi$  及其定义域内一个元素  $i$  满足  $\pi(i) = i$ , 称  $i$  为  $\pi$  的一个不动点

求  $\{1, 2, 3, \dots, n\} \rightarrow \{1, 2, 3, \dots, n\}$  的所有双射函数中, 恰好有  $k$  个不动点的函数的种类数

Hint:  $n$  个点的错排数  $D(n)$  已知

**Answer:**

$k$  个不动点共有  $\binom{n}{k}$  种取法

剩下  $(n - k)$  个点均为错排, 共  $D(n - k)$  种

所以总种类数为  $\binom{n}{k} D(n - k) = \binom{n}{k} \cdot (n - k)! \cdot \sum_{i=0}^{n-k} (-1)^i \frac{1}{i!}$

## Problem 3.4

将  $n$  对夫妇安置在  $2n$  把椅子的圆桌旁, 要求每一对夫妇均不相邻, 求不相同的安排方案的种类数

**Answer:**

不考虑限制条件, 将  $2n$  个人安排在  $2n$  个圆桌位置的种类数为

$$T = (2n - 1)!$$

设  $A_i$  表示事件“第  $i$  对夫妇相邻”, 则所求安排方案种类数为

$$S = T - |A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_n|$$

求  $|A_i|$  相当于把第  $i$  对夫妇看为一个整体, 之后将  $2n - 1$  个元素安排在  $2n - 1$  个位置上; 而在这对夫妇内部有 2 种安排方法, 所以

$$|A_i| = (2n - 2)! \cdot 2$$

同理

$$|A_i \cap A_j| = (2n - 3)! \cdot 2^2$$

根据容斥原理

$$\begin{aligned} S &= T - \left( \sum_i |A_i| - \sum_{i \neq j} |A_i \cap A_j| + \cdots + (-1)^{n+1} |A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_n| \right) \\ &= T - \left[ \binom{n}{1} 2(2n - 2)! - \binom{n}{2} 2^2(2n - 3)! + \cdots + (-1)^{n+1} \binom{n}{n} 2^n(n - 1)! \right] \\ &= \sum_{i=0}^n (-1)^i \binom{n}{i} 2^i (2n - 1 - i)! \end{aligned}$$

## Homework 4

### Problem 4.1

求  $(2 + x)^{\frac{3}{2}} / (1 - x)$  中  $x^3$  的系数

**Answer:**

$$\begin{aligned}\frac{(2+x)^{\frac{3}{2}}}{1-x} &= (2+x)^{\frac{3}{2}}(1+x+x^2+\cdots) \\ &= \sum_{k=0}^{\infty} \binom{\frac{3}{2}}{0} x^k 2^{\frac{3}{2}-k} (1+x+x^2+\cdots)\end{aligned}$$

其中  $x^3$  的系数为

$$\begin{aligned}&\binom{\frac{3}{2}}{0} 2^{\frac{3}{2}} + \binom{\frac{3}{2}}{1} 2^{\frac{1}{2}} + \binom{\frac{3}{2}}{2} 2^{-\frac{1}{2}} \\ &= 2^{\frac{3}{2}} + \frac{3}{2} 2^{\frac{1}{2}} + \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 2^{-\frac{1}{2}} \\ &= \frac{59}{16} \sqrt{2}\end{aligned}$$

## Problem 4.2

求下列序列的生成函数（不能用无穷级数表示）

(1)  $0, 0, 0, 0, -6, 6, -6, 6, \dots$

(2)  $1, 0, 1, 0, 1, 0, \dots$

(3)  $1, 2, 1, 4, 1, 8, \dots$

**Answer:**

(1)

$$\begin{aligned}(1, 1, 1, 1, \dots) &\Leftrightarrow \frac{1}{1-x} \\ (1, -1, 1, -1, \dots) &\Leftrightarrow \frac{1}{1-(-1)x} = \frac{1}{1+x} \\ (1, -1, 1, -1, \dots) &\Leftrightarrow -\frac{1}{1+x} \\ (6, -6, 6, -6, \dots) &\Leftrightarrow -\frac{6}{1+x} \\ (0, 0, 0, 0, 6, -6, 6, -6, \dots) &\Leftrightarrow -\frac{6x^4}{1+x}\end{aligned}$$

(2)

$$\begin{aligned}(1, 1, 1, 1, \dots) &\Leftrightarrow \frac{1}{1-x} \\ (1, -1, 1, -1, \dots) &\Leftrightarrow \frac{1}{1-(-1)x} = \frac{1}{1+x} \\ (2, 0, 2, 0, \dots) &\Leftrightarrow \frac{1}{1-x} + \frac{1}{1+x} \\ (1, 0, 1, 0, \dots) &\Leftrightarrow \frac{1}{2} \left( \frac{1}{1-x} + \frac{1}{1+x} \right)\end{aligned}$$

(3)

$$\begin{aligned}
(1, 0, 1, 0, \dots) &\Leftrightarrow \frac{1}{2} \left( \frac{1}{1-x} + \frac{1}{1+x} \right) \\
(0, 1, 0, 1, \dots) &\Leftrightarrow \frac{x}{2} \left( \frac{1}{1-x} + \frac{1}{1+x} \right) \\
(0, 2, 0, 4, \dots) &\Leftrightarrow (0(\sqrt{2})^1, (\sqrt{2})^2, 0(\sqrt{2})^3, (\sqrt{2})^4, \dots) \\
&\Leftrightarrow \sqrt{2} \cdot \frac{\sqrt{2}x}{2} \left( \frac{1}{1-\sqrt{2}x} + \frac{1}{1+\sqrt{2}x} \right) = x \left( \frac{1}{1-\sqrt{2}x} + \frac{1}{1+\sqrt{2}x} \right) \\
(1, 2, 1, 4, 1, 8, \dots) &\Leftrightarrow \frac{1}{2} \left( \frac{1}{1-x} + \frac{1}{1+x} \right) + x \left( \frac{1}{1-\sqrt{2}x} + \frac{1}{1+\sqrt{2}x} \right)
\end{aligned}$$

## Problem 4.3

求  $i + 3j + 3k = n$ ,  $i \geq 0, j \geq 1, k \geq 1$  的整数解  $< i, j, k >$  的个数

**Answer:**

即求  $(1 + x + x^2 + \dots)(x^3 + x^6 + x^9 + \dots)(x^3 + x^6 + x^9 + \dots)$  中  $x^n$  的系数

$$\begin{aligned}
&(1 + x + x^2 + \dots)(x^3 + x^6 + x^9 + \dots)(x^3 + x^6 + x^9 + \dots) \\
&= \frac{1}{1-x} \cdot \frac{x^3}{1-x^3} \cdot \frac{x^3}{1-x^3} = x^6(1 + x + x^2)(1 - x^3)^{-3}
\end{aligned}$$

所以  $x^n$  的系数为

$$\left( \frac{-3}{\frac{n-6}{3}} \right), \left( \frac{-3}{\frac{n-7}{3}} \right), \left( \frac{-3}{\frac{n-8}{3}} \right)$$

取  $\frac{n-6}{3}, \frac{n-7}{3}, \frac{n-8}{3}$  为整数的一项（易知有且只有一项为整数）

## Problem 4.4

$a(x)$  为  $(a_0, a_1, a_2, \dots)$  的生成函数, 求  $(a_0, a_0 + a_1, a_0 + a_1 + a_2, \dots)$  的生成函数

**Answer:**

$$\begin{aligned}
&a_0 + (a_0 + a_1)x + (a_0 + a_1 + a_2)x^2 + \dots \\
&= a_0(1 + x + x^2 + \dots) \\
&\quad + a_1(x + x^2 + x^3 + \dots) \\
&\quad + a_2(x^2 + x^3 + x^4 + \dots) \\
&\quad + \dots \\
&= \frac{a_0}{1-x} + \frac{a_1x}{1-x} + \frac{a_2x^2}{1-x} + \dots \\
&= \frac{1}{1-x} a(x)
\end{aligned}$$

# Homework 5

## Problem 5.1

求  $a_n$  的非递推形式

$$(1) a_0 = 2, a_1 = 3, a_{n+2} = 3a_n - 2a_{n+1}$$

$$(2) a_0 = 1, a_{n+1} = 2a_n + 3$$

**Answer:**

$$(1) \text{特征方程 } x^2 + 2x - 3 = 0 \text{ 解得 } x_1 = -3, x_2 = 1$$

$$\text{令 } a_n = a(-3)^n + b$$

$$\begin{cases} a + b = 2 \\ -3a + b = 3 \end{cases} \quad \begin{cases} a = -\frac{1}{4} \\ b = \frac{9}{4} \end{cases}$$

$$\therefore a_n = -\frac{1}{4}(-3)^n + \frac{9}{4}$$

$$(2) \text{特征方程 } x = 2 \text{ 齐次解 } a'_n = p \cdot 2^n \text{ 通解 } a_n = p \cdot 2^n + q$$

$$\begin{cases} p + q = 1 \\ 2p + q = 5 \end{cases} \quad \begin{cases} p = 4 \\ q = -3 \end{cases}$$

$$\therefore a_n = 4 \cdot 2^n - 3$$

## Problem 5.2

$$a_{n+2} = \sqrt{a_{n+1}a_n}, a_0 = 2, a_1 = 8, \text{ 求 } a_n \text{ 的解析式及 } n \rightarrow \infty \text{ 时的 } a_n$$

**Answer:**

$$\text{令 } b_n = \log_2 a_n$$

$$\text{则有 } \log_2 a_{n+2} = \frac{1}{2}(\log_2 a_{n+1} + \log_2 a_n)$$

$$\text{即 } 2b_{n+2} = b_{n+1} + b_n$$

$$\text{特征方程 } 2x^2 - x - 1 = 0$$

$$\text{解得 } x_1 = -\frac{1}{2}, x_2 = 1$$

$$\text{设 } b_n = p\left(-\frac{1}{2}\right)^n + q$$

$$\text{根据 } b_0 = 1, b_1 = 3$$

有

$$\begin{cases} p + q = 1 \\ -\frac{1}{2}p + q = 3 \end{cases} \quad \begin{cases} p = -\frac{4}{3} \\ q = \frac{7}{3} \end{cases}$$

$$\therefore b_n = -\frac{4}{3}\left(-\frac{1}{2}\right)^n + \frac{7}{3}, a_n = 2^{-\frac{4}{3}}\left(-\frac{1}{2}\right)^n + \frac{7}{3}, \lim_{n \rightarrow \infty} a_n = 2^{\frac{7}{3}}$$

## Problem 5.3

求证:  $\forall n \geq 0, \frac{1}{2}[(1 + \sqrt{2})^n + (1 - \sqrt{2})^n] \in \mathbb{Z}$

**Answer:**

$$\lambda_1 = 1 + \sqrt{2}, \lambda_2 = 1 - \sqrt{2} \quad \lambda_1 + \lambda_2 = 2, \lambda_1 \lambda_2 = -1$$

构造以  $\lambda_1, \lambda_2$  为根的一元二次方程  $x^2 - 2x - 1 = 0$

$n = 0, 1$  时  $\frac{1}{2}[(1 + \sqrt{2})^n + (1 - \sqrt{2})^n]$  分别为 1, 1

构造数列  $a_0 = 1, a_1 = 1, a_{n+2} = 2a_{n+1} + a_n$  的解析式为  $a_n = \frac{1}{2}[(1 + \sqrt{2})^n + (1 - \sqrt{2})^n]$

由于只涉及正整数的加减乘运算, 所以数列的各项均为整数, 即  $\forall n \geq 0, \frac{1}{2}[(1 + \sqrt{2})^n + (1 - \sqrt{2})^n] \in \mathbb{Z}$

## Problem 5.4

计算

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} (-4)^{-k}$$

**Answer:**

$$\text{令 } f(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} (-4)^{-k}$$

$$\begin{aligned} f(n) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} (-4)^{-k} \\ &= 1 + \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n-k}{k} (-4)^{-k} \\ &= 1 + \sum_{k=1}^{\lfloor n/2 \rfloor} \left[ \binom{n-1-k}{k} + \binom{n-1-k}{k-1} \right] (-4)^{-k} \\ &= \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-1-k}{k} (-4)^{-k} + \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n-1-k}{k-1} (-4)^{-k} \\ &= \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-1-k}{k} (-4)^{-k} + \sum_{k=0}^{\lfloor (n-2)/2 \rfloor} \binom{n-2-k}{k} (-4)^{-(k+1)} \\ &= f(n-1) - \frac{1}{4}f(n-2) \end{aligned}$$

特征函数  $x^2 - x + \frac{1}{4} = 0$  解得  $x_1 = x_2 = \frac{1}{2}$

$$\text{设 } f(n) = \frac{an+b}{2^n}$$

$$\begin{cases} f(0) = b & = 1 \\ f(1) = \frac{1}{2}(a+1) & = 1 \end{cases} \quad \begin{cases} a = 1 \\ b = 1 \end{cases}$$

$$\therefore f(n) = \frac{n+1}{2^n}$$

## Homework 6

### Problem 6.1

给定下列  $f(n), g(n)$ , 判断  $f(n) = O(g(n)), f(n) = \Omega(g(n))$  是否成立

$$(1) f(n) = 2n^3 + 3n, g(n) = 100n^2 + 2n + 100$$

$$(2) f(n) = 50n + \log n, g(n) = 10n + \log \log n$$

$$(3) f(n) = 50n \log n, g(n) = 10n \log \log n$$

$$(4) f(n) = \log n, g(n) = \log^2 n$$

$$(5) f(n) = n!, g(n) = 5^n$$

**Answer:**

$$(1) f(n) = \Omega(g(n))$$

$$(2) f(n) = O(g(n)), f(n) = \Omega(g(n))$$

$$(3) f(n) = \Omega(g(n))$$

$$(4) f(n) = O(g(n))$$

$$(5) f(n) = \Omega(g(n))$$

### Problem 6.2

举例增函数  $f(n), g(n)$ , 使得  $f(n) \neq O(g(n)), f(n) \neq \Omega(g(n))$

**Answer:**

$$f(n) = n^{n+\sin n}, g(n) = n^{n+\cos n}$$

### Problem 6.3

按照增长速度排序以下函数

$$(1)n \ln n \quad (2)(\ln \ln n)^{\ln n} \quad (3)(\ln n)^{\ln \ln n}$$

$$(4)n \cdot e^{\sqrt{\ln n}} \quad (5)(\ln n)^{\ln n} \quad (6)n \cdot 2^{\ln \ln n}$$

$$(7)n^{1+\frac{1}{\ln \ln n}} \quad (8)n^{1+\frac{1}{\ln n}} \quad (9)n^2$$

**Answer:**

取对数

$$(1) \ln n + \ln \ln n$$

$$(2) \ln n \cdot \ln \ln \ln n$$

$$(3) \ln \ln n \cdot \ln \ln n$$

$$(4) \ln n + \sqrt{\ln n}$$

$$(5) \ln n \cdot \ln \ln n$$

$$(6) \ln n \cdot \ln 2 \ln \ln n$$

$$(7) \left(1 + \frac{1}{\ln \ln n}\right) \ln n = \ln n + \frac{\ln n}{\ln \ln n}$$

$$(8) \left(1 + \frac{1}{\ln n}\right) \ln n = \ln n + 1$$

$$(9) 2 \ln n$$

借鉴快速排序的思路，以(1)为 pivot

$$(1)(2) \ln n \cdot \ln \ln \ln n \succ 2 \ln n \succ \ln n + \ln \ln n \quad (2) \succ (1)$$

$$(1)(3) x^2 \prec e^x \therefore (\ln \ln n)^2 \prec e^{\ln \ln n} = \ln n \prec \ln n + \ln \ln n \quad (3) \prec (1)$$

$$(1)(4) \sqrt{x} \succ \ln x \therefore \sqrt{\ln n} \succ \ln \ln n \quad (4) \succ (1)$$

$$(1)(5) \ln n \cdot \ln \ln n \succ 2 \ln n \succ \ln n + \ln \ln n \quad (5) \succ (1)$$

$$(1)(6) \ln 2 \cdot \ln \ln n \prec \ln \ln n \quad (6) \prec (1)$$

$$(1)(7) \ln n = e^{\ln \ln n} \succ (\ln \ln n)^2 \therefore \frac{\ln n}{\ln \ln n} \succ \ln \ln n \quad (7) \succ (1)$$

$$(1)(8) 1 \prec \ln \ln n \quad (8) \prec (1)$$

$$(1)(9) \ln n \succ \ln \ln n \quad (9) \succ (1)$$

$$\therefore (3)(6)(8) \prec (1) \prec (2)(4)(5)(7)(9)$$

(3)(6)(8): 以(3)为 pivot

$$(3)(6) \ln n \succ \ln \ln n \quad (6) \succ (3)$$

$$(3)(8) \ln n \succ (\ln \ln n)^2 \quad (8) \succ (3)$$

$$(6)(8) 1 \prec \ln 2 \ln \ln n \quad (8) \prec (6)$$

$$\therefore (3) \prec (8) \prec (6)$$

(2)(4)(5)(7)(9): 以(2)为 pivot

$$(2)(4) \ln n \cdot \ln \ln \ln n \succ 2 \ln n \succ \ln n + \sqrt{\ln n} \quad (2) \succ (4)$$

$$(2)(5) \ln \ln \ln n \prec \ln \ln n \quad (2) \prec (5)$$

$$(2)(7) \ln n \cdot \ln \ln \ln n \succ 2 \ln n \succ \ln n + \frac{\ln n}{\ln \ln n} \quad (2) \succ (7)$$

$$(2)(9) \ln n \cdot \ln \ln \ln n \succ 2 \ln n \quad (2) \succ (9)$$

$$\therefore (4)(7)(9) \prec (2) \prec (5)$$

(4)(7)(9): 以(4)为 pivot

$$(4)(7) \sqrt{\ln n} \succ \ln \ln n \therefore \frac{\sqrt{\ln n}}{\ln \ln n} \succ 1 \therefore \frac{\ln n}{\ln \ln n} \succ \sqrt{\ln n} \quad (7) \succ (4)$$

$$(4)(9) \ln n \succ \sqrt{\ln n} \quad (9) \succ (4)$$

$$(7)(9) \ln n \succ \frac{\ln n}{\ln \ln n} \quad (9) \succ (7)$$

$\therefore (4) \prec (7) \prec (9)$

最终排序(3)(8)(6)(1)(4)(7)(9)(2)(5)

## Problem 6.4

若  $f_1(n) = O(g_1(n))$ ,  $f_2(n) = O(g_2(n))$ ,

证明  $f_1(n) + f_2(n) = O(g_1(n) + g_2(n))$ ,  $f_1(n)f_2(n) = O(g_1(n)g_2(n))$

**Answer:**

根据定义有

$$\forall n > N_1, f_1(n) \leq c_1 g_1(n)$$

$$\forall n > N_2, f_2(n) \leq c_2 g_2(n)$$

所以

$$\forall n > \max\{N_1, N_2\}, f_1(n) + f_2(n) \leq c_1 g_1(n) + c_2 g_2(n) \leq \max\{c_1, c_2\}(g_1(n) + g_2(n))$$

$$\forall n > \max\{N_1, N_2\}, c = c_1 c_2, f_1(n)f_2(n) \leq c_1 c_2 g_1(n)g_2(n) = c g_1(n)g_2(n)$$

即证  $f_1(n) + f_2(n) = O(g_1(n) + g_2(n))$ ,  $f_1(n)f_2(n) = O(g_1(n)g_2(n))$

## Problem 6.5

证明:  $2\sqrt{n+1} - 2 < \sum_{i=1}^n \frac{1}{\sqrt{i}} < 2\sqrt{n} - 1$

**Answer:**

$$\sum_{i=1}^n \frac{1}{\sqrt{i}} > \int_1^{n+1} \frac{1}{\sqrt{x}} dx = 2\sqrt{x} \Big|_1^{n+1} = 2\sqrt{n+1} - 2$$
$$\sum_{i=1}^n \frac{1}{\sqrt{i}} = 1 + \sum_{i=2}^n \frac{1}{\sqrt{i}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx = 1 + 2\sqrt{x} \Big|_1^n = 2\sqrt{n} - 1$$